K Rithwik Rag Goud

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.76
SRI CHAITANYA JUNIOR COLLEGE, DD COLONY, HYDERABAD	Class XII (SSC)	2023	96.90%
SRI CHAITANYA HIGH SCHOOL, GADDIANNARAM,HYDERABAD	Class X (SSC)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

- **AlgoUniversity Technology Fellowship (ATF) 2025** - Qualified Stage 1, ranked in the **top 8% nationwide** among 250,000 participants; advanced to Stage 2.

SKILLS

- **Skills:** Data Structures & Algorithms
- **Tools & Platforms**: VS Code, Git, GitHub.
- **Languages** : C, C++, Python, SQL.
- **Databases**: MySQL.

RELEVANT COURSES

- **Mathematics**: Probability, Applied Statistics, Linear Algebra, Convex Optimization, Transform Techniques, Intro to Metric Spaces, Number Theory, Ordinary Differential Equations, Calculus I and II.
- **Computer Science**: Introduction to Programming, Data Structures and Applications, Algorithms, Database Management Systems I, Theory of Computation, Compilers I.

PROJECTS

- **AdTech Campaign Performance & Product Analytics |** SQL, MySQL (view project)
- Analyzed **1,143+ advertising records** across impressions, clicks, spend, and conversions to evaluate campaign and funnel performance.
- Used **advanced SQL, window functions, and audience segmentation** to compare campaign performance across age, gender, and interest segments.
- Evaluated **CTR, CPC, conversion rate, and CPA** to identify performance gaps and generate data-driven opportunities for campaign optimization.

- **Statistical Analysis of Real-World Datasets** | *Course Project* |*Python, Pandas, NumPy, SciPy, Statsmodels, Matplotlib*(View Report)
- Performed **one- and two-tailed hypothesis testing** on real-world datasets using t-statistics, p-values, and confidence-based inference.
- Applied **chi-square testing and contingency tables** to investigate the relationship between categorical variables.
- Built an **linear regression model** using Scikit-Learn, evaluating coefficient significance via **t-statistics/p-values** and model fit with **R^2**.
- Used **Pandas, NumPy, Scikit-Learn, SciPy, and Matplotlib** for data preprocessing, regression modeling, statistical hypothesis testing, and visualization.

- **LZ77 Compression Engine using Suffix Tree** | C++, STL(View Project)
- Implemented the **LZ77 lossless text compression algorithm** in C++ using a suffix-tree-based data structure for efficient longest-match search.
- Designed a modular architecture with separate components for compression logic, suffix tree construction, and file processing.
- Encoded input text into (**offset, length, next character**) tokens while supporting configurable search and look-ahead windows.
- Applied string processing and custom data structures without relying on external compression libraries.

POSITIONS OF RESPONSIBILITY

- **Core Team Member - Hospitality,Abhyant** | IIT Hyderabad
- Coordinated accommodation, food, and hospitality arrangements for participants, guests, and teams at IIT Hyderabad's annual sports fest.

EXTRACURRICULAR

- **Quantitative Problem Solving:** Achieved an elite **2000+ rating across all formats on Matiks**(top 100 globally across multiple formats).
- **Competitive Chess:** Represented at the regional level with a **1700 rating on Chess.com.**
- **Typing Speed:** Achieved a peak touch typing speed of **100 WPM on Monkeytype.**

- As part of **National Service Scheme(NSS)**, participated in community awareness campaigns and walkathons promoting health, social responsibility and public welfare.